BAHRIA UNIVERSITY ISLAMABAD

Department of Computer Science

Assignment 1

Subject: MAD

BS-IT 6

Submission Date = As on LMS

Dear Students,

Please choose your app idea and, **as a group of two or three students**, submit a **project proposal** for approval. The project will be completed in **three stages** as outlined below:

Assignment 1 – Project Proposal

Submit a clear and well-structured proposal for your proposed mobile application. The proposal must contain the following details:

1. App Title & Description

- Provide a clear and meaningful name for your app.
- Write a short description explaining the main purpose and functionality of the application.

2. Problem Statement

- Clearly identify the problem or need that your application is intended to address.

3. Project Objectives

- List 3–5 main objectives of the proposed application.
- Explain what the application is intended to achieve.

4. Target Users

- Specify who will use the application, such as students, teachers, customers, patients, employees, etc.

5. Reference / Justification

- If the app is not based on a client or client–server model, provide a reference source to justify your idea.
- Alternatively, you may choose an existing application from the Google Play Store and include its Play Store link.
- Clearly mention any external application, tutorial, API, dataset, or other source used as a reference.

6. Main Features

- List the major features that will be implemented in the application.
- Examples: registration/login, add/update/delete/view data, search/filter, notifications, profile management, etc.

7. App Flow Screens – Basic Sketch or Wireframe

- Show the expected flow of the application in terms of screens.
- Example: Splash → Login → Dashboard → Profile.
- Use simple diagrams, hand-drawn sketches, or digital mockups.
- Show how users move from one screen to another using arrows/navigation flow.

8. Technical Requirements

- Mention the programming language and framework to be used.
- Mention development tools such as Flutter, Dart, Android Studio, or VS Code.
- Mention APIs or other technologies if required.

9. Database Requirement

- The application must use either local storage or a cloud database.
- Examples of local storage: SQLite, SharedPreferences, Hive, etc.
- Examples of cloud databases: Firebase Realtime Database, Firebase Firestore, etc.

10. Expected Database Structure

- Provide a basic database design.
- Mention proposed collection/table names and important fields.
- Show relationships between data where applicable.

12. Scope & Limitations

- Clearly mention what the application will cover.
- Mention features or functionality that are outside the scope of the current project.

13. Future Enhancements

- List 2–3 features that could be added in future versions but are not required for the current project.

14. Proposal Approval

- Students must receive instructor approval before starting Assignment 2.
- Any major change in the approved app idea, features, database, or technology must be discussed with and approved by the instructor.

Assignment 2 – UI Submission

Submit only the User Interface (UI) of your proposed application.

- Submit UI code only; no backend/database functionality is required at this stage.
- Design all major screens included in the approved proposal.
- Implement navigation between screens.
- Use proper colors, typography, spacing, alignment, icons, and images.
- Ensure the UI is user-friendly and responsive.
- Submit screenshots of the completed UI.
- Submit the complete source code/project.

Assignment 3 – Complete App Submission

Submit the final functional version of the application with both UI and backend fully implemented.

- Complete UI implementation.
- Database/backend implementation.
- CRUD operations where applicable.
- Authentication if included in the approved proposal.
- API integration if required.
- Complete navigation and workflow according to the approved proposal.
- Data validation and appropriate error handling.
- Functional testing of the application.
- Submit the final APK.
- Submit the complete source code/project.

Assignment 4 – Presentation

Present your completed application and explain the design and implementation.

- App introduction
- Problem being solved
- Project objectives
- Target users
- Main features
- UI design
- App navigation/flow
- Database/backend implementation
- Demonstration of the working application
- Challenges faced and solutions
- Future enhancements
- Conclusion

Submission Note

Make sure your proposal is clear, well-structured, and contains all required references, app flow screens, features, database details, technical requirements, scope, and future enhancements. The final application should follow the approved proposal.